

Lecheheb Djaafar — IoT & Automation Specialist — Oil & Gas Industry Expert

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Available for relocation and international assignments

Professional Summary

Innovative Software Engineer with specialization in IoT solutions and industrial automation for the oil and gas sector. Demonstrated expertise in developing robust monitoring systems that enhance production efficiency and reliability. Combines technical knowledge with practical field experience in major petroleum facilities. Skilled in Python development, data analytics, and implementing real-time visualization solutions that drive operational excellence.

Professional Experience

Major Petroleum Companies

Algeria

IoT Systems Engineering Intern

2023–2024

- Engineered and deployed IoT-based monitoring systems that increased production visibility by 35%.
- Developed automated data collection protocols that reduced manual reporting time by 65%.
- Implemented ThingsBoard dashboards delivering real-time telemetry visualization for critical oil production parameters.
- Collaborated with field engineers to integrate sensor networks in challenging oilfield environments.

Freelance

Remote

Python Solutions Developer

2022–2025

- Architected custom automation solutions for industrial clients, increasing operational efficiency by 25%.
- Developed scalable web applications using Django and Flask frameworks with RESTful API integration.
- Created data processing pipelines that transformed raw sensor data into actionable business intelligence.
- Provided technical consultation for cloud infrastructure migration and optimization.

Education

Qasdi Merbah University

Ouargla, Algeria

Bachelor's Degree in Computer Science

2023

Specialization: Software Development and IoT Systems

Key Coursework: Embedded Systems Programming, Industrial Automation, Data Analytics, Network Security

Technical Expertise

Programming Languages: Python (Advanced), C/C++ (Intermediate), Java (Intermediate), JavaScript (Proficient)

IoT & Automation: ThingsBoard, MQTT Protocol, Raspberry Pi, ESP32, Arduino, Industrial Sensors, PLC Integration

DevOps & Tools: Docker, Linux Administration, Git Version Control, CI/CD Pipelines, AWS Cloud Services

Databases: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, Firebase), Time-Series Databases (InfluxDB)

Web Development: Django, Flask, React.js, RESTful API Design, Bootstrap, Responsive Design

Data Analysis: Pandas, NumPy, Data Visualization (Matplotlib, Plotly), Jupyter Notebooks

Professional Certifications

2024: Advanced Industrial IoT Implementation - ThingsBoard Professional
2023: Python for Data Science and Machine Learning - Certified Developer
2023: Introduction to Artificial Intelligence and Machine Learning
2022: Oil & Gas Production Technology - Fundamentals Certification

Field Experience

Hassi Messaoud (Houd El Hamra)

Algeria

Technical Field Internship

2024

- Received comprehensive training on well testing procedures and production systems.
- Collaborated with senior engineers on implementing sensor networks for production monitoring.
- Participated in daily operations meetings and troubleshooting sessions.

Berkawi Oil Field

Algeria

Production Operations Internship

2023

- Gained hands-on experience in oil production operations and equipment.
- Assisted in data collection and analysis for production optimization.
- Developed understanding of safety protocols and operational procedures.

Languages

Arabic: Native

Fluent in both written and spoken forms

English: Professional Working Proficiency

Technical documentation and communication

French: Intermediate

Business conversation capable

Notable Projects

Remote Monitoring System: Designed and implemented an end-to-end IoT solution for remote well monitoring using ESP32 microcontrollers, MQTT protocol, and custom dashboards.

Predictive Maintenance App: Developed a machine learning model to predict equipment failures in pumping systems, reducing unplanned downtime by 20%.

Production Reporting Tool: Created an automated reporting system that consolidated data from multiple sources into comprehensive daily production reports.

Professional Attributes

Leadership: Led technical teams on multiple projects, demonstrating strong coordination and delegation skills

Problem Solving: Analytical approach to troubleshooting complex technical and operational challenges

Adaptability: Proven ability to quickly learn new technologies and adapt to changing project requirements

Communication: Excellent technical documentation skills and ability to explain complex concepts to non-technical stakeholders

Additional Information

Availability: Immediately available for opportunities in the oil and gas sector globally

Interests: Chess strategy, Emerging technology trends, Contributing to open-source IoT projects

Professional Memberships: Society of Petroleum Engineers (SPE) - Student Member